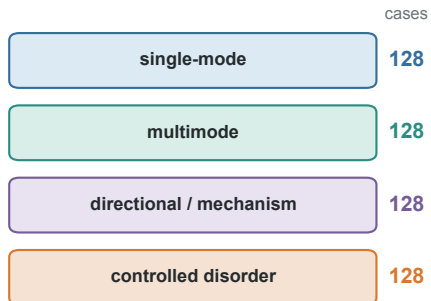
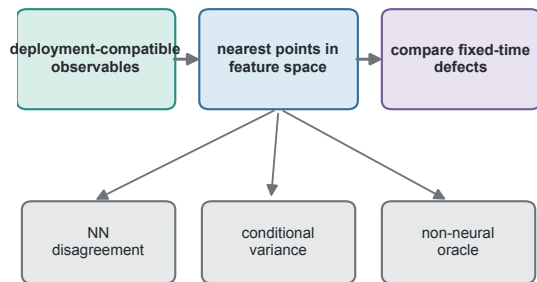


a Mechanism-stratified development atlas



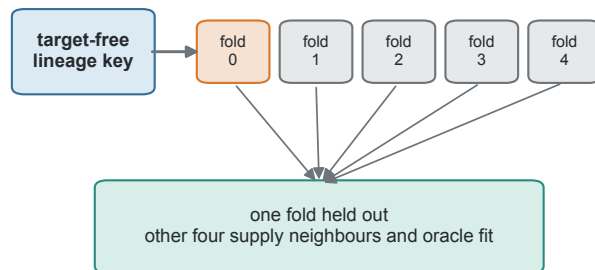
512 complete cases; balanced before target evaluation

c Observable-to-ambiguity diagnostic



Coverage is assessed independently from conditional ambiguity

b Field-lineage-held-out diagnostic folds



No particle-level split; cases and folds receive equal weight

d Initial operational-identifiability summary

Density rate	P	N	P	P	P	N	P
Pressure gradient	P	N	N	N	N	N	P
Viscosity Laplacian	P	N	P	N	N	N	P
	NN median	NN P90	Cvar upper 95%	Oracle NRMSE	Oracle improvement	Worst-family NRMSE	Coverage

P/N labels preserve the decision encoding in grayscale